Semantic Representation Fusion-Based Network for Robust Land Cover Classification in Foggy Conditions

Weipeng Shi, Wenhu Qin, Zhonghua Yun, Chao Wu, Yukun Yang, and Tao Zhao

***Abstract*— A precise and robust classification of land cover is crucial for land use estimation. A robust model that can provide rich semantic information is imperative for the challenging task of land cover classification (LCC) in foggy conditions. We propose semantic representation enhancement (SRE) and semantic representation aggregation (SRA) modules for the fusion of semantic representation. The dense depthwise separable atrous spatial pyramid pooling (DDS-ASPP) module in SRE possesses a large receptive field, which covers an extensive scale range. An enhanced asymmetric convolution module (EACM) in SRE focuses on features of various directions. DDS-ASPP and EACM generate the class-based and pixel-based representation, respectively. By means of SRA and dual representations, we model the relationship between global context and coarse class regions to capture long-range correlation. Moreover, evaluated on Potsdam, Vaihingen, and custom real-world datasets under fog, we demonstrate that our work is competitive with state-of-the-art models in terms of robustness. Code will be available at https://github.com/bowenroom/Robust-land-cover-classification.**

***Index Terms*— Attention mechanism, land cover classification (LCC), low light condition, robust deep learning, semantic segmentation.**

NOMENICLATURE

AAP	Adaptive average pooling layer.
EACM	Enhanced asymmetric convolution module.
CNN	Convolutional neural network.
DDS-ASPP	Dense depthwise separable atrous spatial pyramid pooling.
FCN	Fully convolutional network.
GSD	Ground sampling distance.
HRNet	High-resolution net.
LCC	Land cover classification.
mCD	Mean corruption degradation.
NSIs	Natural scene images.
rCD	Relative corruption degradation.
RSIs	Remote sensing images.
SRE	Semantic representation enhancement.
SRA	Semantic representation aggregation.

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I. INTRODUCTION

SEMANTIC labeling (a.k.a. semantic segmentation) of RSIs refers to the pixel-level fine-grained LCC, allowing for the efficient measurement of land use and change detection. Deep learning in recent years has motivated the development of remote sensing to a great extent. A number of studies have attempted to build applications based on this technique, such as data fusion [1], image decomposition [2], earth observation [3], image classification [4], and so on. Basically, numerous academics have investigated into semantic segmentation based on NSIs, such as datasets, such as BDD100k, Apollo scape, Cityscape, and so on. In contrast, there is few literatures devoted exclusively to robust remote sensing under fog. Unlike NSIs, RSIs with fog exhibit a variety of complex characteristics that may impair the segmentation robustness. First is the issue of scale diversity. The first column in Fig. 1 illustrates the variation in vehicle sizes caused by fog occlusion and tree shading. It is possible for objects of the same class to differ in size and shapes. In the dense prediction task of RSIs, CNN is not always robust when sizes vary drastically. Second is the issue of low image quality. In dense fog, quality of RSIs is generally poor, and this will significantly degrade the classification accuracy. Low interclass differentiation is accompanied by the high intraclass variance in RSIs. The second column in Fig. 1 shows impervious surfaces made of various materials. Although objects in dashed boxes pertain to the same class, they are visually distinct. This is homogeneous heterogeneity. In addition, objects from heterogeneous classes will be mistakenly regarded as the identical when they share close features, such as materials and colors. In particular, when objects in RSIs are engulfed in dense fog, few distinct features may be available. This is heterogeneous homogeneity.

Primary aim of this article is to propose a reliable framework, which is robust with regard to fog corruptions. To cope with the listed challenges, we incorporate the proposed SRE and SRA into HRNet [5]. HRNet consists of parallel connected multiresolution branches and maintains the high-resolution

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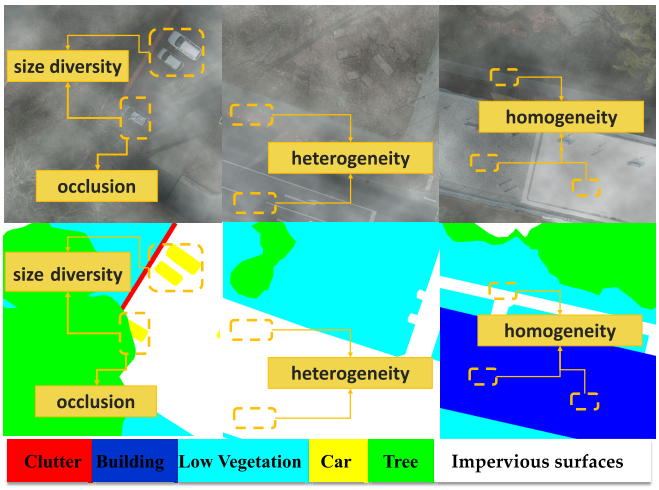


Fig. 1. Challenges of LCC under fog.

representation during the whole procedure. This facilitates the strong robustness in fog. SRE is composed of DDS-ASPP and EACM. DDS-ASPP combines the depthwise separable and dilated convolution with spatial pyramid pooling to classify multiscale land covers properly at a lower computing expense. EACM, which focuses on features of various directions, substitutes the commonly used 3×3 layer with 1×3, 3×1, 3×3, and diagonal convolution kernels. Conventional successive downsampling operation leads to the inherent loss of beneficial clues, which affects LCC accuracy. Consequently, it is tough for CNN to differentiate between targets and surrounding pixels due to the lack of distinct features. Therefore, we get hold of the association between adjacent pixels and soft object regions with the aid of attention mechanism. SRA refines the coarse class-based representations to capture long-range dependencies with global feature maps, which are essential for the enrichment of semantic information. Hence, we can improve LCC robustness by probing into multiscale context with the fusion of SRE and SRA. Overall, major contributions are described as follows.

1) For evaluating classification robustness in the real world, we collected land cover data in Nanjing, Jiangsu, China, on both clean and foggy days using an unmanned aerial vehicle (UAV). A dataset with annotation is released for model training and robustness testing. We conduct ablation studies on the customized dataset to verify the effectiveness.
2) Based on HRNet, we propose and integrate SRE and SRA to build an end-to-end model for the robust semantic segmentation of RSIs.
3) The proposed SRE improves the multiscale representation capability.
4) The proposed SRA can capture the correlation between pixels and coarse class regions over a long distance, enhancing the classification robustness when dealing with RSIs of low quality.
5) Through ablation studies and comparisons on foggy datasets (including both synthetic and real data), we show that our model achieves promising performance for robust LCC under fog.

II. Related Works

In this section, we will briefly describe the related work involved in this article, including semantic segmentation, attention mechanism, and some works about robustness.

Semantic segmentation is usually constructed based on two basic attributes of gray values: similarity and discontinuity. Gray-scale mutations are used to detect the discontinuous edge information, and a predefined criterion is adopted to segment the original into approximate subregions. Deep learning-based models have significantly advanced over approaches that manually extract features, such as points, lines, and edges. There are three types of deep learning models for semantic segmentation [5]: The first type are models, such as FCN [6] and [7], which learns semantic representation at low resolution by associating the coarse feature maps to specific pixel points. Similar structures are DeepLab [8], pyramid scene parsing network (PSPNet) [9], DilatedNet [10], dense atrous spatial pyramid pooling (DenseASPP) [11], and so on. Downsampling in the encoder of this structure relies on cascading pooling, which inherently reduces the sensitivity to relationship between various spatial locations. It is, therefore, not feasible to guarantee the accuracy and robustness for each category as localization effectiveness diminishes. The second are networks, such as UNet [12], which downsample the original image and then gradually upsample from the low-resolution representation via an encoder–decoder structure akin to SegNet [13] Hourglass [14], RefineNet [15], DeepLabv3+ [16], and so on. Global information can be taken into account as well as the local details. Nevertheless, it is less effective when RSIs contain indiscernible object with diverse shapes and sizes. A third group consists of models, such as HRNet [5], which exchange feature information by concatenating branches at each resolution in parallel, thus maintaining high-resolution representation at all times. Another structure is GridNet [17]. Using HRNet, multiresolution representations are repeated in order to render rich, high-resolution representations that are highly position sensitive. This improves the robustness by a large margin.

Nonlocal [18] operations enable models to obtain long-range dependencies. An example of this is self-attention, which takes the interaction between various parts into consideration regardless of the element location. Dual attention network (DANet) [19] applies self-attention to fuse local features with global dependencies, as well as to filter feature information according to different channels. Criss-cross network (CCNet) [20] brings forward a novel crossover attention, which can guarantee both computational efficiency and accuracy. Object-contextual representation network (OCRNet) [21] utilizes attention to explore the relationship between objects and surrounding pixel context. Similar network structures involve double-attention network (A^2Net) [22]. PSANet [23] put forward an adaptive learning attention mask to attenuate the restrictive influence between surrounding pixels. Asymmetric non-local neural network (ANNN) [24] constructs an asymmetric nonlocal and fusion structure, which can substantially improve the precision and performance simultaneously. While these models rely on self-attention mechanism

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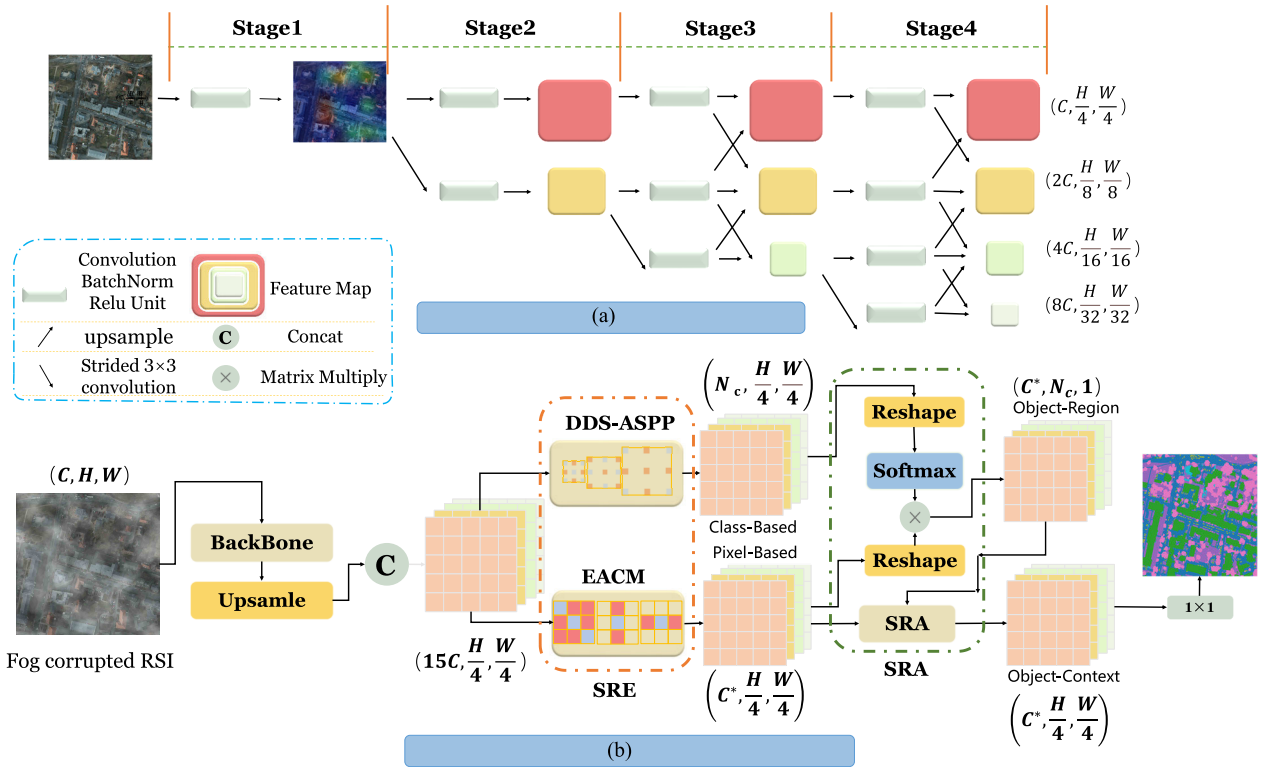


Fig. 2. Overview of the overall pipeline architecture. SRE and SRA are denoted by orange and green dashed boxes, respectively. SRE can encode multiscale semantic representations. SRA captures the dependency between pixels and instances through the pixel-based and class-region-based representations, thereby enhancing robustness. (a) Backbone structure. (b) Pipeline of the proposed network.

to improve classification accuracy, they also introduce a large number of parameters, resulting in redundant noise within the model. Foggy conditions, furthermore, result in poor robustness for classifying corrupted RSIs, because they do not generalize well.

Thus far, the vast majority of studies on semantic segmentation focuses on the classification accuracy instead of robustness. Whereas, robustness is notably necessary for the daily application. Le Saux and Boujemaa [25] proposed a robust unsupervised and adaptive clustering method for image database categorization. Hendrycks and Dietterich [26] set up benchmarks for testing the robustness of classifier and introduced new datasets IMAGENET-C and IMAGE-P. Kamann and Rother [27] conducted the comparison study of semantic segmentation robustness on autonomous driving datasets. The majority of these studies investigate how to improve the perceptive robustness for natural scenes. RSIs with regard to fog corruption possess unique characteristics, and the classification robustness cannot be guaranteed by just applying conventional deep convolutional networks. Our paper concentrates on developing robust modules and evaluating them using rendered and real-world data to improve performance.

III. Method

A. Overview

An overview of the architecture is illustrated in Fig. 2. HRNet is selected as the backbone. SRE consists of DDS-ASPP and EACM to jointly enhance the classification robustness. The proposed modules generate feature maps, which are related to coarsely extract class regions and pixel context, respectively. Subsequently, by fusing the dependency between distant pixel points and object regions, SRA improves the overall classification accuracy and robustness.

B. Backbone

Fine-grained semantic segmentation of tiny objects engulfed in fog requires a robust model, which is sensitive to spatial locations and long-range dependency. High-resolution representations contain superior local details, whereas low-resolution representations are marked by global information and powerful classifying abilities, which complement one another. As a backbone, we prefer HRNet, because it can maintain high-resolution representations, which focus on various fog covered regions throughout the entire process to avoid performance degradation. By fusing representations from multiple receptive fields, HRNet is characterized with excellent semantic expressiveness and robustness to RSIs of low quality.

As shown in Fig. 2, HRNet [5] consists of parallel connected multiresolution branches (C, $2C$, $4C$, and $8C$ in Fig. 2). Each branch can be divided into four stages. Between neighboring stages, there exists a multiresolution fusion module (represented by cross rows, which are stride convolution and upsampling operations in Fig. 2). This facilitates the flow of information between multiple branches. LCC is associated with semantic segmentation of RSIs, so we adopt the structure, such as HRNet-W48. Detailed specification about HRNet architecture is illustrated in Table I. Module in each stage is depicted in a way, such as $[a \times a, n\ C] \times b \times c$. C represents channel numbers, and "[]" is the residual unit. Parameters

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TABLE I
DETAILED HRNET STRUCTURE PARAMETERS ABOUT EACH STAGE AND NUMBER OF OUTPUT CHANNELS

Downsp. rate	Stage1	Stage2	Stage3	Stage4
4×	$\begin{bmatrix} 1\times1, 64 \\ 3\times3, 64 \\ 1\times1, 256 \end{bmatrix}\times4\times1$	$\begin{bmatrix} 3\times3, C \\ 3\times3, C \end{bmatrix}\times4\times1$	$\begin{bmatrix} 3\times3, C \\ 3\times3, C \end{bmatrix}\times4\times1$	$\begin{bmatrix} 3\times3, C \\ 3\times3, C \end{bmatrix}\times4\times3$
8×		$\begin{bmatrix} 3\times3, 2C \\ 3\times3, 2C \end{bmatrix}\times4\times1$	$\begin{bmatrix} 3\times3, 2C \\ 3\times3, 2C \end{bmatrix}\times4\times4$	$\begin{bmatrix} 3\times3, 2C \\ 3\times3, 2C \end{bmatrix}\times4\times3$
16 ×			$\begin{bmatrix} 3\times3, 4C \\ 3\times3, 4C \end{bmatrix}\times4\times4$	$\begin{bmatrix} 3\times3, 4C \\ 3\times3, 4C \end{bmatrix}\times4\times3$
32 ×				$\begin{bmatrix} 3\times3, 8C \\ 3\times3, 8C \end{bmatrix}\times4\times3$



(a) (b) (c)

Fig. 3. Cars, which are covered by different severity levels of fog, could vary in sizes and shapes. Diverse geometric sizes impair the model robustness. (a) Severity level 5. (b) Severity level 3. (c) Severity level 1.

a–c are kernel size, duplication times of the residual unit, and entire modularized part, respectively. $\begin{bmatrix} 3\times3, nC \\ 3\times3, nC \end{bmatrix}$ is the basic block. Four modularized residual units accompanied by the fusion element compose each branch, which correspond to different resolutions.

C. SRE Module

In the study of LCC under fog, instances often exhibit multiscale characteristics, which can adversely impact the classification robustness. First, significant disparities in scale can be observed between different classes. For instance, the notable size difference between the white car and the building in Fig. 3(a) illustrates this point. Second, objects of the same class may undergo substantial size variations when obscured by dense fog or trees. An example is the cars positioned at various locations in Fig. 3(b), which demonstrate these alterations in scales. Therefore, we designed SRE to enhance multiscale representation capability. DDS-ASPP retains a relatively large receptive field to encode multiscale representation in a dense way. EACM, which is robust to features of various directions in RSIs, is utilized to produce pixel-based relational context. DDS-ASPP extracts class-based feature map coarsely as well (N_c in Fig. 2: number of classes), which refers to class-centered regions of RSIs. From EACM, we can acquire the relationship between different locations, which will facilitate the subsequent representation fusion.

1) DDS-ASPP: LCC under fog especially requires joint interpretation of the multiscale context. By continuously downsampling and gradually recovering from the low-resolution representation, encoder–decoder models can accurately capture the outer edge of land covers. Nonetheless, loss of spatial details is inevitable during the process of dimension reduction. Meanwhile, objects in fog vary in sizes. DCNN alone for semantic segmentation would not bring the improvement in robustness by a large margin. As Fig. 3 shows, car shapes and sizes appear diverse due to the uncertainty introduced by fog.

Inspired by Kamann and Rother [27], we can conclude that ASPP with dilated convolution can enhance the robustness in a variety of image impairment situations, such as natural noise, light change, and so on. Dilated convolution is characterized by a better integration of surrounding contextual information. It can well enlarge the receptive field to capture objects of various sizes in RSIs. The accumulation of atrous layers increases the size of receptive field exponentially. DDS-ASPP takes advantage of densely connected depthwise separable atrous convolution and spatial pyramidal pooling. Comparing with ASPP, which concatenate dilation layers in parallel, DDS-ASPP is featured with denser pixel and scale sampling as well as the larger receptive field, which is robust for the classification of tiny objects in RSIs. In addition, with the merit of depthwise separable convolution, DDS-ASPP can get a denser pixels sampling in case of the decimation of detail signal with a low computing cost. Thus, we exploit DDS-ASPP to address the scale variation challenges with a large receptive field on the one hand, and, on the other hand, acquire soft class-centered regions roughly to calibrate the information fusion.

Architecture of DDS-ASPP is illustrated in Fig. 4. DDS-ASPP densely concatenates the original input feature map with a 3 × 3 depthwise separable convolution with five dilation rates (3, 6, 12, 18, and 24) to obtain the representation from different scales. The joint layers can generate multiscale representations via 1 × 1 convolution. The channel dimension of the output feature map equals the number of classes. DDS-ASPP covers a wide range of scales while being more cost-effective. It is capable of achieving a balance between accuracy and efficiency in the classification of multiscale objects.

Equation (1) shows how dilated convolution applied on the input signal $\boldsymbol{x}/\boldsymbol{X}$, where $\boldsymbol{O}$ indicates the output feature map, $\boldsymbol{i}$ denotes the currently retrieved location, $\boldsymbol{d}, \boldsymbol{s}$ denotes the expansion coefficient (dilation rate) and stride, respectively, $\boldsymbol{p}$ is the padding size, $\boldsymbol{K}$ represents the size of the filter, and $\boldsymbol{w}[\boldsymbol{k}]$ is the kth filter. The depthwise separable convolution consists of a depthwise convolution and a pointwise convolution [28]. When $\boldsymbol{d} = \boldsymbol{1}$, we can get the convolution operator as usual. $\boldsymbol{X}_{\text{size}}, \boldsymbol{O}_{\text{size}}$ are the size of input as well as output feature map correspondingly

$$O[i] = \sum_{k=1}^{K} x[i + d \cdot k]w[k] \tag{1}$$

$$O_{\text{size}} = \left[\frac{X_{\text{size}} + 2p - k - (k-1)(d-1)}{s}\right] + 1 \tag{2}$$

$$R_d = (d-1) \times (K-1) + K \tag{3}$$

$$R_{([d_1,d_2,\ldots,d_J],J)} = R_{([d_1,d_2,\ldots,d_{J-1}],J-1)} + \left(k^* - 1\right) \times \prod_{j=1}^{J-1} \left(s_j\right). \tag{4}$$

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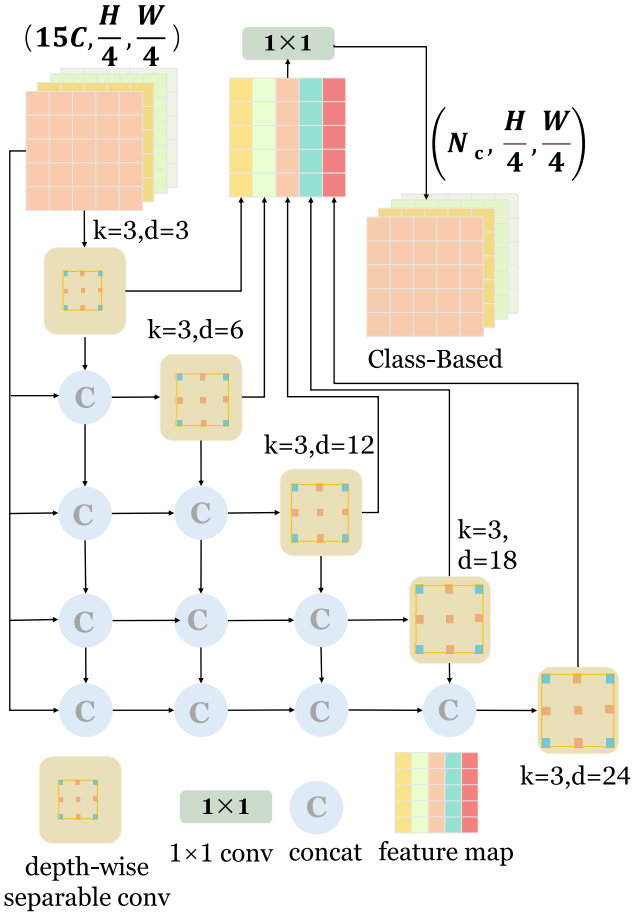


Fig. 4. Architecture of DDS-ASPP. It can be applied to RSIs for robust and accurate encoding of multiscale context. The output feature map is associated with class regions. (k: kernel size and d: dilation rate).

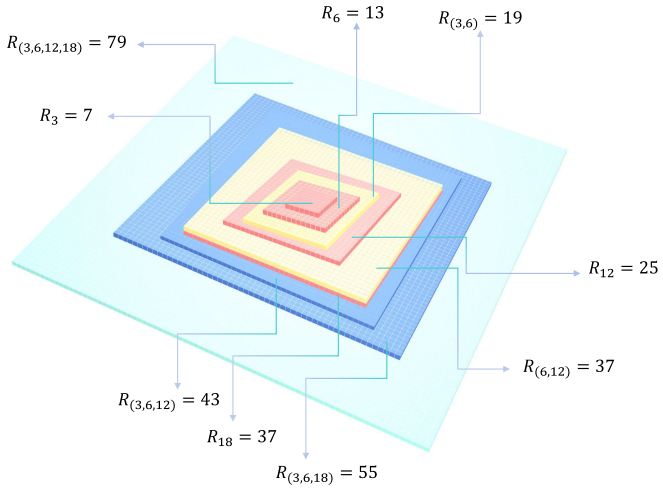


Fig. 5. Illustration of pyramid structure for densely connected ASPP layers with varying dilation rates (3, 6, 12, and 18). For example, $R_{(3,6,18)} = 55$ signifies that after stacking three layers with the dilation rates of 3, 6, and 18, the size of receptive field (or the equivalent kernel size k^*) is 55. Different colors represent stacking different numbers of dilated convolution layers.

In (2) and (3), $\boldsymbol{R}$ means the size of receptive field, $\boldsymbol{J}$ is the index number of cascading layers, and $[d_1, d_2, \ldots, d_J]$ denotes stacking $\boldsymbol{J}$ layers whose dilation rates are $\boldsymbol{d}$. $\boldsymbol{k^*}$ is the matching kernel size, which is equal to $\boldsymbol{R_d}$ (not stacking other convolution layers). $\boldsymbol{s_j}$ represents stride of layer $\boldsymbol{j}$. Let us take the scale pyramid (shown in Fig. 5) as a running example. We set stride for all layers as 1, and the size of the convolving kernel $\boldsymbol{K}$ is 3. For example, $R_6 = 5 \times 2 + 3 = 13$, and receptive filed size of layer (dilation rate $= 6$) is 13. $R_{(6,12)}$ means stacking two layers in cascade whose dilation rates are 6 and 12 and $R_{([6,12],2)} = R_{(6,1)} + (R_{12} - 1) \times 1 = 13 + [(12 - 1) \times (3 - 1) + 3 - 1] \times 1 = 37$.

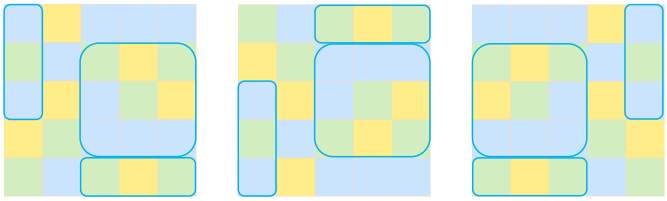


Fig. 6. Square kernel extracts flipped feature map, which is distinct from the original one. However, 3×1 and 1×3 kernels will keep identical. EACM focuses on the feature map from different directions, which is robust to rotational distortions.

2) EACM: Conventional convolution kernels handle pixels differently, with attention weights usually higher at the cross center and lower at the corners. Motivated by Ding et al. [29] and Liu et al. [30], we suggest integrating EACM into SRE to concentrate on visual patterns from various directions, thereby increasing robustness. Upon implementing image augmentation through vertical flipping, it can be observed that the feature maps derived from square convolution kernels undergo alterations. However, when employing horizontal and vertical convolution kernels, the extracted feature is identical as the original at axis-symmetric positions, as shown in Fig. 6. EACM is depicted in Fig. 7. EACM can generate a feature map associated with pixel representation without extra computational load during the inference stage.

EACM is proposed based on the basic convolution feature of *equivalent kernel* [29]. For **2-*D*** convolution, we can get the equivalent kernel size by summing up different kernels on the corresponding position. Details can be expressed by (5), where $\boldsymbol{M}$ and $\boldsymbol{K}$ signify matrix and kernel, respectively [(1)–(5) mean different but compatible sizes]. $\oplus$ is the elementwise summation. EACM attaches small convolution kernels to the larger one. In the training stage, the original 3×3 convolutional layers are substituted with the fused sum of five convolutions of 3×1, 1×3, 3×3, and two diagonal convolutions as the output of convolutional layers

$$\boldsymbol{M} * K^{(1)} + \boldsymbol{M} * K^{(2)} + \cdots + \boldsymbol{M} * K^{(5)} = \boldsymbol{M} * \left(K^{(1)} \oplus K^{(2)} \cdots \oplus K^{(5)}\right). \quad (5)$$

D. SRA Module

In Fig. 3(c), the car enclosed within the rectangular box exhibits a resemblance to the building. A robust model must rely on the contextual relationship and surrounding pixels to differentiate them. To obtain semantic association between long-range pixels without bringing excess parameters, we develop SRA, which can model long-range dependency without heavy computation burden. Leveraging pixel-based and class-based representations, SRA capitalizes

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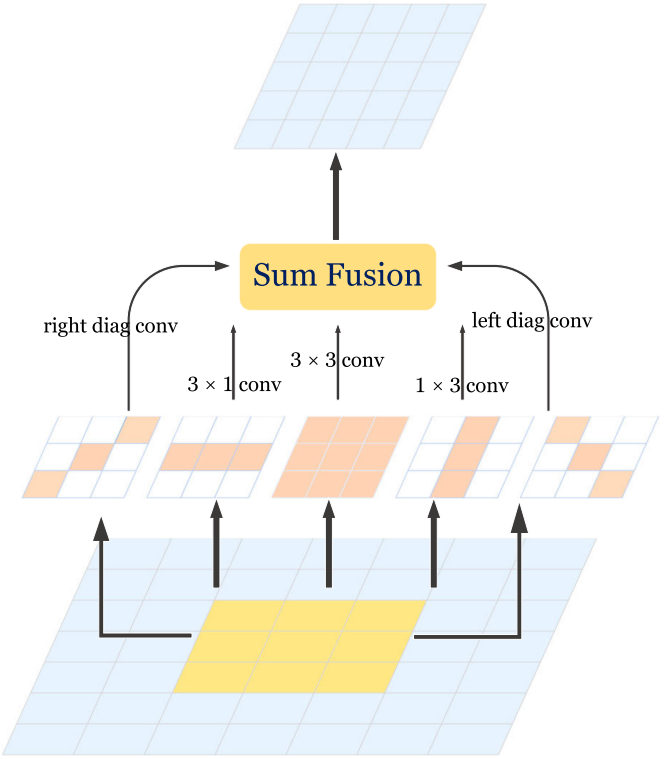


Fig. 7. Architecture of EACM. EACM substitutes the original 3×3 convolution with five types of convolution kernels.

on the relationship between instances and their adjacent pixels. According to the conclusion from global context network (GCNet) [31], there is little distinction between the obtained attention weights and query points; i.e., query positions and global semantic context do not appear to be strongly associated. Thus, we exclude the query part.

Detailed structure is shown in Fig. 8. We obtain the long-range dependency between pixels from EACM output (pixel-based feature map in Fig. 8, denoting as X_{pixel}, $X_{\text{pixel}} \in \mathcal{R}^{C^* \times (H/4) \times (W/4)}$). Fusing the transformed X_{pixel}, we acquire the representation X_B under the coarse class-based representation X_{class} ($\boldsymbol{X}_{\text{class}} \in \mathcal{R}^{C^* \times \mathcal{N}_c \times 1}$). We can use (6)–(10) to depict this procedure

$$X_A = \text{AAP}\left(X_{\text{pixel}}\right) \tag{6}$$

$$X_B = X_{\text{class}} \times \boldsymbol{\rho}\left(X_{\text{pixel}}\right) \tag{7}$$

$$O_c = X_{\text{pixel}} + W_{v2}\left(F_{LN}\left(F_{\text{Mish}}\left(W_{v1}\sum_{j=1}^{N_p}\frac{e^{W_k X_{Aj}}}{\sum_{m=1}^{N_p} e^{W_k X_{Am}}} X_{Bj}\right)\right)\right) \tag{8}$$

$$F_{\text{Mish}}(x) = x \cdot \tanh\left(\ln\left(1 + e^x\right)\right) \tag{9}$$

$$F_{LN}(x) = \frac{x - E[x]}{\sqrt{\text{Var}[x] + \epsilon}} * \gamma + \beta. \tag{10}$$

We utilize an AAP layer to squeeze the global information $\boldsymbol{X}_{\text{pixel}}$ to a specified shape size for global context modeling. ρ denotes a series of transformation procedures, including key embedding, softmax, and reshaping of the tensor. Key embedding layer (weight as $\boldsymbol{W_k}$) also enhances the nonlinearity for capturing the long-range dependencies. After the matrix multiplication of $\boldsymbol{X}_{\text{class}}$ and transformed $\boldsymbol{X}_{\text{pixel}}$, we can get

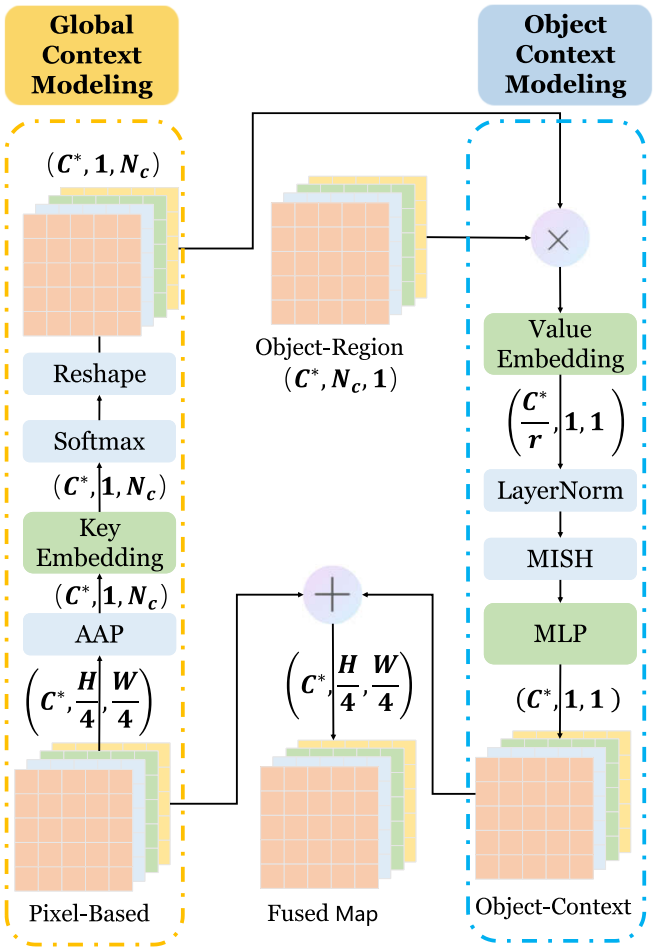


Fig. 8. Architecture of SRA. We apply SRA to model the long-range relationship efficiently and enrich semantic features.

the feature map $\boldsymbol{X_B}$, $\boldsymbol{X_B} \in \mathcal{R}^{C^* \times 1 \times 1}$. This step is to model the context. Next, we fuse the coarse class-based feature maps, while using the value embedding layer for channel dimension compression (weight as $\boldsymbol{W_{v1}}$), thus reducing the parameter amount ($\boldsymbol{F}_{\text{LN}}$ and $\boldsymbol{F}_{\text{Mish}}$ are LayerNorm [32] and Mish activation [33], respectively). MLP layer is appended as a bottleneck transform (weight as $\boldsymbol{W_{v2}}$). $\boldsymbol{\gamma}$ and $\boldsymbol{\beta}$ in F_{LN} are trainable vectors, which is for affine transformation and ϵ is for numerical stability. So that we can explore channel-wise dependency relationships. Finally, semantic features are enriched by sum fusion with X_{pixel}.

IV. Experiment

We present a general introduction about experiment datasets in this section first. Then, we explain how to preprocess and generate fog corrupted test set. Finally, the evaluation metrics and experiment details are discussed.

A. Dataset Overview

To assess the classification performance under real-world foggy conditions, we acquired clean and foggy RSIs of natural environments using a UAV (Fig. 9). Some sample images from our dataset are shown in Fig. 10. Data were collected from 22 unique locations in Nanjing, Jiangsu, China, as denoted by the black points in Fig. 11. Throughout the data collection

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Fig. 9. UAV for data collection, DJI Mavic3, equipped with a Hasselblad camera with 20 million effective pixels.

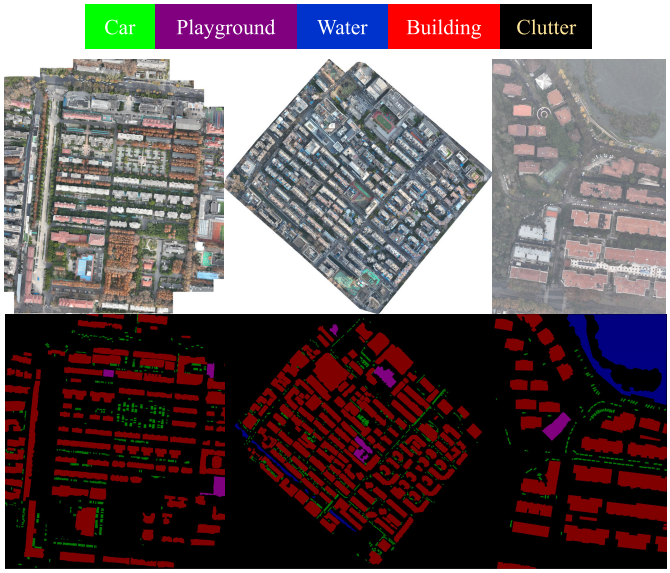


Fig. 10. Overview of some images and labels from our Nanjing dataset. Images captured in fog, such as the rightmost, are utilized to assess the robustness performance.

process, we maintained a flight altitude of 200 m and a GSD of 3 cm. We generated orthophotos and manually annotated each image according to the following categories: clutter, car, playground, water, and building. We employed clear RSIs for model training and foggy data for testing. This approach allowed us to evaluate the extent of performance degradation in the presence of fog.

For the comparison with other SOTA, we also use Potsdam and Vaihingen datasets of International Society for Photogrammetry and Remote Sensing (ISPRS) [34]. Sampled at 900-m altitude, these datasets for rural villages and complex cities have respective resolutions of 6000 × 6000 and 2500 × 2000. GSDs are 5 and 9 cm. Each dataset contains six classes, such as cars, low vegetation, trees, buildings, impervious surfaces, and clutter background. The original images of both datasets are shown in Fig. 12.

To streamline data processing and modeling, we conduct an exploratory data analysis of the ISPRS dataset. By extracting the composition of all pixels from the ground truth (GT), we present the analysis results in Fig. 13. The proportion of cars is the smallest, representing less than 2% of the total pixels. Conversely, the proportions of other classes exhibit relatively minor variations.

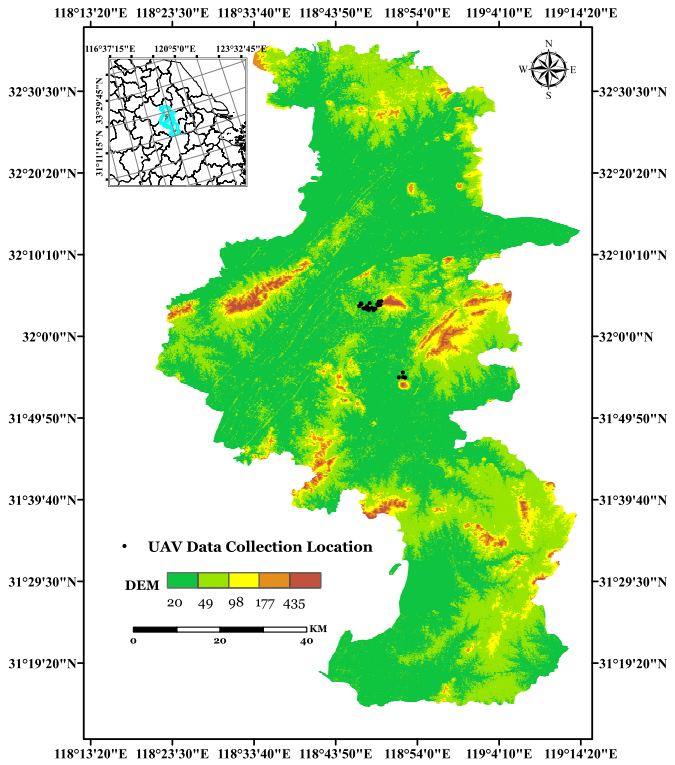


Fig. 11. Locations of data collection by UAV and digital elevation model (DEM) information. (Nanjing, Jiangsu, China).

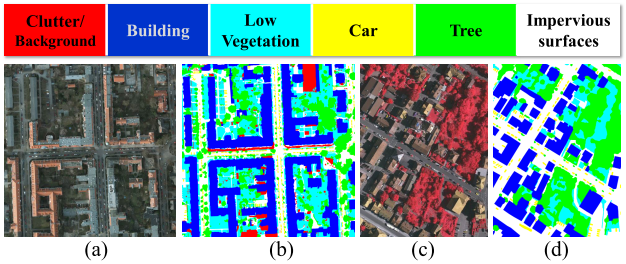


Fig. 12. Overview of original images and labels in Vaihingen and Potsdam datasets. (a) Potsdam image. (b) Potsdam label. (c) Vaihingen image. (d) Vaihingen label.

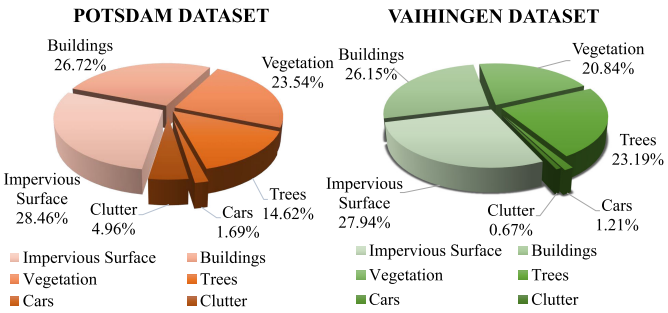


Fig. 13. Illustrations about the composition of each class in ISPRS dataset.

B. Preprocessing and Augmentation

Each RSI in Potsdam and Vaihingen has a resolution of 6000 × 6000 and average 2000 × 2000, respectively. Limited by GPU memory, we need to crop all the original images to 512 × 512. Because of a label error, image with id **4–12**

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Fig. 14. Overview of the original RSIs along with the augmented images after the random extraction and transformation.

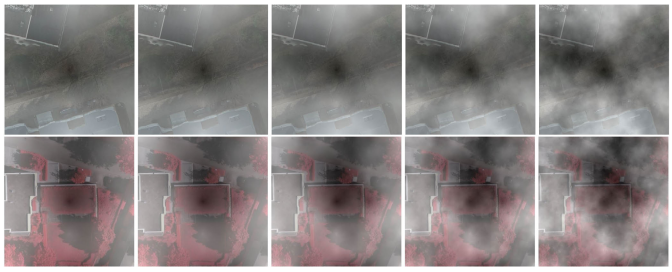


Fig. 15. Illustration of five severity levels of fog corruption. First row: Potsdam. Second row: Vaihingen. Progressing from left to right, the corruption severity incrementally intensifies.

in Potsdam has been removed. We set the slicing stride size to 128. Due to the relatively small amount of the training set, we also conduct data augmentation. We use a variety of augmentation methods, including random patch extraction, resizing, rotating, erasing, and adjusting the brightness, contrast, saturation, and hue. Augmented training set samples are illustrated in Fig. 14.

ISPRS[1] offers RSIs, which were captured in normal meteorological conditions. With a view to verify the robustness, we need to observe the adaptability in several corrupted conditions. Test set augmentation is required. According to [26], we render five different degrees of fog corruption through diamond-square algorithm. Test set with various levels of fog corruption is illustrated in Fig. 15. Performance degradation for all models is evaluated across a range of severity levels.

C. Evaluation Metrics

To verify effectiveness and robustness for LCC of RSIs, we choose the following metrics for judging the performance, which are overall accuracy (OA), intersection over union (IoU), mean IoU (mIoU), mCD, and rCD. Specifically, precision and recall are defined as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}} \quad \text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{11}$$

where TP, FP, TN, and FN are the number of true positive, false positive, true negative, and false negative, respectively. OA is formulated as follows:

$$\text{OA} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{FP}+\text{TN}+\text{FN}}. \tag{12}$$

IoU is defined as (13), and the higher one indicates a more accurate classification result

$$\text{IoU} = \frac{\text{TP}}{\text{FP}+\text{FN}+\text{TP}}. \tag{13}$$

$\boldsymbol{D}$ in (14) is the performance degradation. Metrics, such as mCD and rCD [26], [27] in (14), are adopted to compare the classification robustness of RSIs across different severity degrees. **mCD** measures the absolute robustness. **rCD** includes the metric measured on a clean dataset. It is a relative ratio value of absolute performance degradation comparing classification scores in the clean and foggy conditions

$$D = 1 - \text{IoU} \tag{14}$$

$$\text{mCD}_F^f = \frac{\widetilde{D_{s,F}^f}}{\widetilde{D_{s,F}^{\text{ref}}}} \times 100\% \tag{15}$$

$$\text{rCD}_F^f = \frac{\widetilde{D_{s,F}^f} - D_{\text{clean}}^f}{\widetilde{D_{s,F}^{\text{ref}}} - D_{\text{clean}}^{\text{ref}}} \times 100\%. \tag{16}$$

Since models differ in the performance, to compare mutually, we choose a baseline model as a reference in the ablation and comparison experiments, which is denoted by ref. In (15) and (16), $\boldsymbol{S}$ means to get the average metric value under different levels of fog. $\boldsymbol{f}$ is the semantic segmentation model used for pixel labeling. Clean and $\boldsymbol{F}$ in (15) and (16) imply clean and fog corrupted RSIs, respectively. For example, $\widetilde{D_{s,f}^F}$ signifies the degradation value, which is acquired by averaging the inference result of $\boldsymbol{f}$ on fog corrupted conditions.

Compared with the reference, mCD $<$ 100% represents a more robust model, and the part below 100% indicates the performance improvement. rCD $<$ 100% means that the degradation value between fog and clean is smaller than the corresponding reference value. Comparing with the reference, mCD and rCD $>$ 100% signify that the model is less robust and lower values stand for stronger robustness with less performance degradation.

D. Implementation Details

We divide the original dataset into training, validation, and test sets according to the ratio of 60%, 20%, and 20%. Our validation set in Potsdam uses images with IDs 2_12, 3_10, 3_11, 3_12, 5_12, 6_12, and 7_12, and our test set uses images with IDs 2_13, 2_14, 3_13, 3_14, 4_13, 4_14, and 4_15. In Vaihingen, images with region numbers area3, area5, area7, area11, area13, area15, and area17 are used as the validation set, and areas numbered area2, area4, area6, area8, area10, and area12 are used as the test set. Other RSIs are utilized as the training set. We crop the high-resolution image into

[1]Potsdam and Vaihingen datasets can be acquired from *https://www.isprs.org/education/benchmarks/UrbanSemLab/2d-sem-label-potsdam.aspx* and *https://www.isprs.org/education/benchmarks/UrbanSemLab/2d-sem-label-vaihingen.aspx* (*Accessed in September, 2022*).

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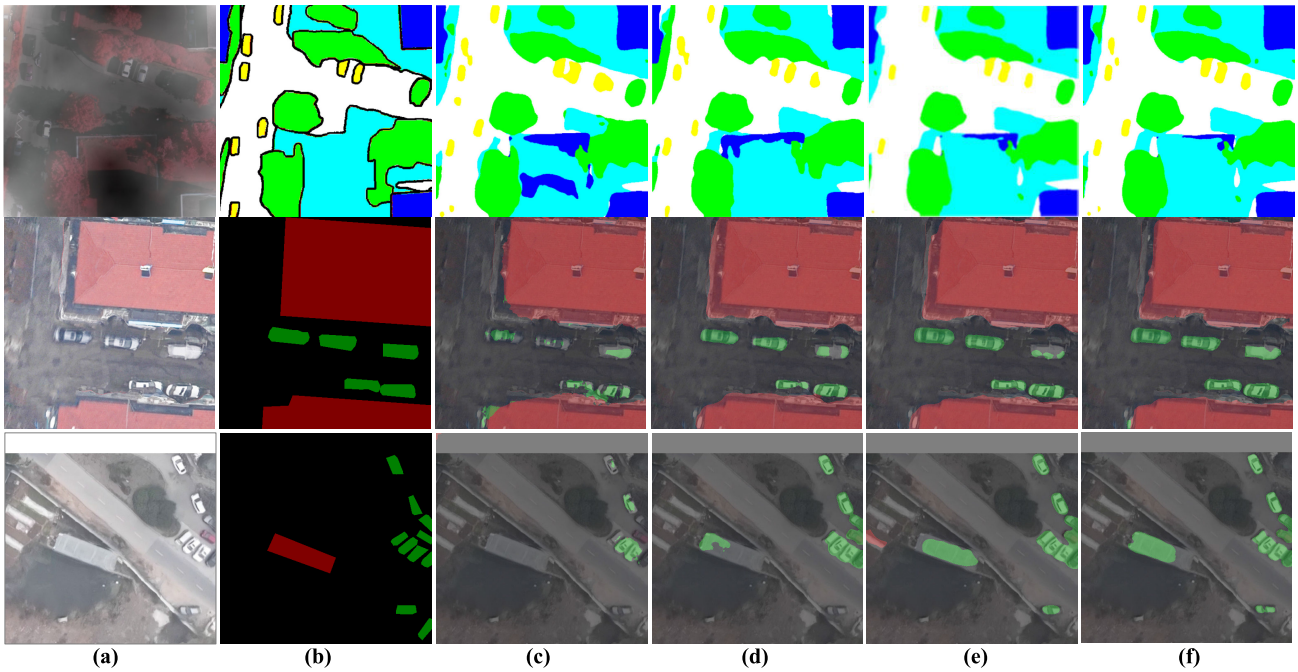


Fig. 16. First row corresponds to the experiment conducted on the Vaihingen dataset, while the rest of the rows depict experiments performed on our self-constructed Nanjing dataset. (a) Image. (b) GT. (c) Baseline. (d) +SRE. (e) +SRA. (f) Whole.

512 × 512 patches and then implement data augmentation in the training set and validation set.

In the process of model training, the backbone HRNetV2-W48 load pretrained weights on ImageNet with parameters of layers, depth, and channels set according to [5]. Since the appropriate learning rate and momentum values are crucial for the final results, we employ one cycle [35] to determine their initial values, which are selected to be 0.00006 and 0.9. A polynomial schedule is applied by default with a factor of 1.0 to decay the learning rate. We use stochastic gradient descent optimizer for training for 20k iterations. Weight decay is $2e-4$. Batch size is searched for using torch memory-adaptive algorithm, which first doubles the initial value until an out of memory (OOM) error is encountered. After then, a binary search algorithm is applied to fine-tune the value. Batch size is set to 6. We also apply the half-precision method provided in [35] to reduce memory consumption. We use focal loss to reduce the class imbalance issue. Therefore, the network can concentrate on training the hard samples while reducing the loss for the easy. For the class *impervious surface, buildings, vegetation, trees, clutter, textand cars* in Potsdam and Vaihingen, we set the weight as [1, 1.07, 1.21, 1.95, 5.74, 16.84] and [1, 1.07, 1.34, 1.21, 41.70, 23.09], respectively. γ in focal loss function for calculating the modulating factor is 2. α value is 0.25 for balancing the importance of positive and negative samples.

V. Result

A. SRE and SRA

1) Visualization Analysis: In order to evaluate and visualize the performance of each component, we add SRE and SRA separately and fuse them together afterward to observe the classification result. The existence of fog degrades the image quality. The baseline model in Fig. 16(c) struggles to accurately segment instance edges in RSIs, as many artifacts are present. Meanwhile, low vegetation is often misclassified as buildings roof. *+SRE/+SRA* means adding SRE and SRA separately, while *whole* stands for incorporating both together to the baseline. From Fig. 16(d), we can observe that SRE improves the multiscale representation ability, since it can classify multiscale instances, even cars with varying orientations, but the grasp of global context is still slightly insufficient. Car edges are clearly visible, but there is still some noise generated. For example, some cars in the last row of Fig. 16 are overlooked due to their colors being similar to the ground. As seen in Fig. 16(e), SRA captures the dependency relationship between different feature channels. However, there are instances where certain details are missing, such as some cars being omitted and trees being misidentified as low vegetation. In contrast to models with a single integrated element, Fig. 16 demonstrates that the whole model is more robust, as there is minimal degradation in the classification result between clean and corrupted conditions. Cars can be clearly classified, and the number of false classifications is reduced. Therefore, it can be concluded that our model can enhance the robustness by effectively acquiring long-range dependencies and extracting dense representations.

2) Quantitative Analysis: We also conduct a quantitative ablation study on Vaihingen. The baseline we choose as a reference is HRNetV2-W48, and pretrained weights on ImageNet are loaded. In the ablation study, we incorporate DDS-ASPP and EACM in SRE as well as SRA separately. When SRA is not integrated, we replace the corresponding input branch with a nonlocal [18] structure, which is composed of key, query, and value components. When DDS-ASPP in SRE is used alone, we substitute EACM with a 3×3 convolution. When utilizing

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TABLE II
ABLATION STUDY RESULT SHOWS THE PERFORMANCE OF EACH COMPONENT INTEGRATED FOR THE CLEAN AND FOG CORRUPTED VAIHINGEN TEST SET

Method			GFLOPS	mIoU(%)		mCD(%)
SRE						
DDS-ASPP	EACM	SRA		Clean	Fog	Fog
baseline			28.88	69.88	60.35	100.00
✓			39.02	70.96	62.01	95.82
	✓		52.42	71.87	63.19	96.89
		✓	52.87	71.91	63.22	92.77
✓	✓		55.60	72.39	63.30	92.57
✓	✓	✓	**66.64**	**73.92**	**65.58**	**86.82**

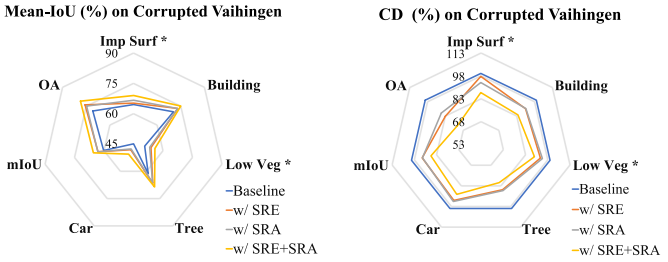


Fig. 17. Two radar plots are presented to demonstrate the results of the ablation study. The CD radar plots, with smaller envelopes, indicate a higher level of robustness.

EACM alone, DDS-ASPP is replaced with a 1×1 convolution. The number of feature channels is identical to the output classes. In Table II, ✓ represents the corresponding module that we incorporate into the baseline. **Clean** represents that the evaluation process is conducted on clean test set, whereas **fog** on fog-covered RSIs acquires the average value across different severity levels. Giga-floating point operations per second (GFLOPS) is calculated with regard to the input size of 512×512.

In Fig. 17, the smaller range of CD radar envelope indicates a more robust model with less performance degradation compared with the reference. From the experimental results, we can find that compared with baseline reference, fusion of SRA and SRE has improved robustness with mCD and rCD generally below 100%. Compared with baseline performance on the clean dataset, the fused model has a minimal performance loss (mCD = 81.03% and rCD = 72.26%) in the classification of trees. Lower values of both verify the robustness. Our model does not show the superior classification performance for the buildings yet. rCD (rCD = 95.13%) is not the least compared with the others. This can be attributed to the regular information in Vaihingen dataset and relatively small shape and texture bias. SRE brings the performance improvement on the clean dataset with mIoU increasing 2.351%. SRA enhances the robustness with mCD and rCD decreasing by 7.23% and 8.75%, respectively, comparing with the reference. According to the results, mCD of the model with SRA is 4.94% lower and IoU is 3.26% higher than the baseline for cars classification in fog.

Effect of DDS-ASPP: Although DDS-ASPP resulted in only a modest 1.0% improvement in performance on clean data, it yielded a more substantial 1.66% increase in classification accuracy in foggy environments and a 4.18% decrease in mCD. The significant improvement in performance in challenging conditions is attributed to the large receptive field and dense feature pyramid for covering a large-scale range. With regard to GFLOPS, the computational complexity increased by 35.11% compared with the baseline model.

Effect of EACM: EACM can further enhance the performance. As shown in Table II, when EACM is added, the classification ability in foggy environments improves by 2.84%, and mCD decreases by 3.11%. Although the performance on clean dataset is similar to DDS-ASPP, its robustness is slightly inferior.

Effect of SRE: After incorporating DDS-ASPP and EACM to construct SRE, the semantic representation capability is significantly improved. The performance on the pristine dataset exhibits a modest enhancement, with a 2.95% increase in mIoU relative to the integration of a solitary module. In contrast, mCD in foggy conditions demonstrates a substantial decline of 7.43%. When compared with the addition of a standalone DDS-ASPP, there is a 42.49% GFLOPS increment, leading to a 3.39% amelioration in the robustness of SRE.

Effect of SRA: Fig. 18 presents the classification performance concerning various degrees of fog corruption. It is evident that low image quality substantially impairs the classification performance. Furthermore, Fig. 18 reveals that our model can still accurately label the majority of buildings when the fog corruption reaches the fourth level. However, under the fifth level, a large portion of building and vegetation are misclassified to a significant degree. Notably, some cars can still be correctly identified. For the baseline model, SRA proves to be highly advantageous in mitigating the impact of natural noise, as evidenced by the significant reduction in mCD by 7.23% after integration. Similar to other modules, the individual performance on the clean dataset shows minimal variation. GFLOPS of SRA and EACM are close.

By integrating the modules described above, our architecture can leverage a large receptive field for classification in scenarios with multiscale objects (e.g., cars and trees with varying sizes). Furthermore, SRA can handle long-range dependencies in RSIs more accurately by combining the coarse class-based and pixel-based representations. Compared with the clean dataset, our model enhances the performance by 5.23% and lowers the mCD by 13.18%. Thus, SRE and SRA improve the classification robustness to fog corruption.

B. Detailed Study on DDS-ASPP

Fig. 19 illustrates the classification performance in foggy scenes with and without DDS-ASPP. As shown in the left and right groups of Fig. 19, which present results on the Potsdam and Nanjing datasets, respectively, model frequently fails to detect cars or misclassifies cars as buildings when DDS-ASPP is not utilized. This confusion is particularly prone to occur when cars are parked under the trees, buildings, or dense fog. By leveraging sufficient semantic context from large scenes, DDS-ASPP can thereby effectively handle the multiscale issues that arise in foggy RSIs. The result showcases its ability to encode multiscale representations, which is beneficial to the robust scene parsing.

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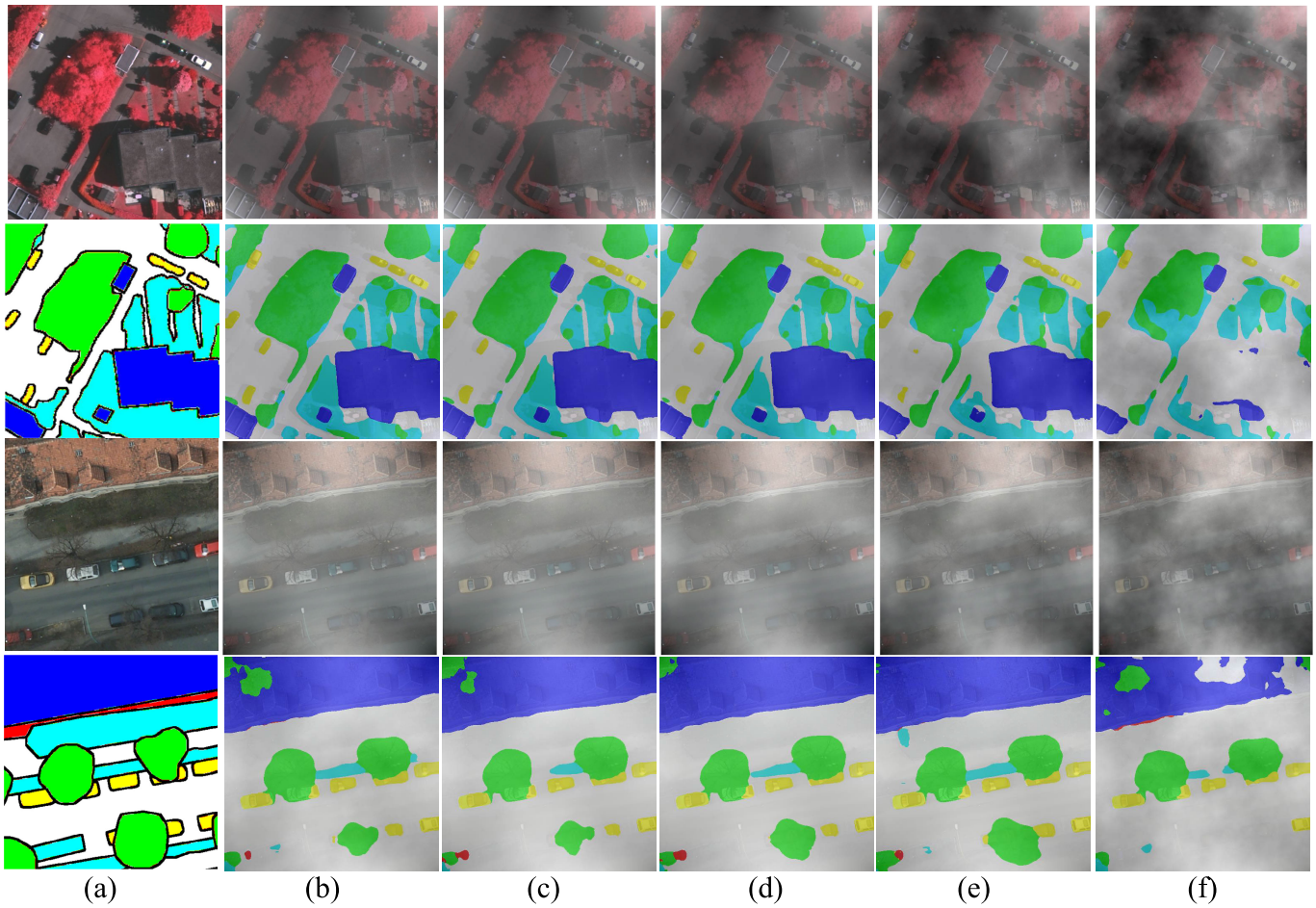


Fig. 18. Classification performance on Vaihingen and Potsdam datasets with regard to different levels of fog corruption. (a) Clean and GT. (b) Level 1. (c) Level 2. (d) Level 3. (e) Level 4. (f) Level 5.

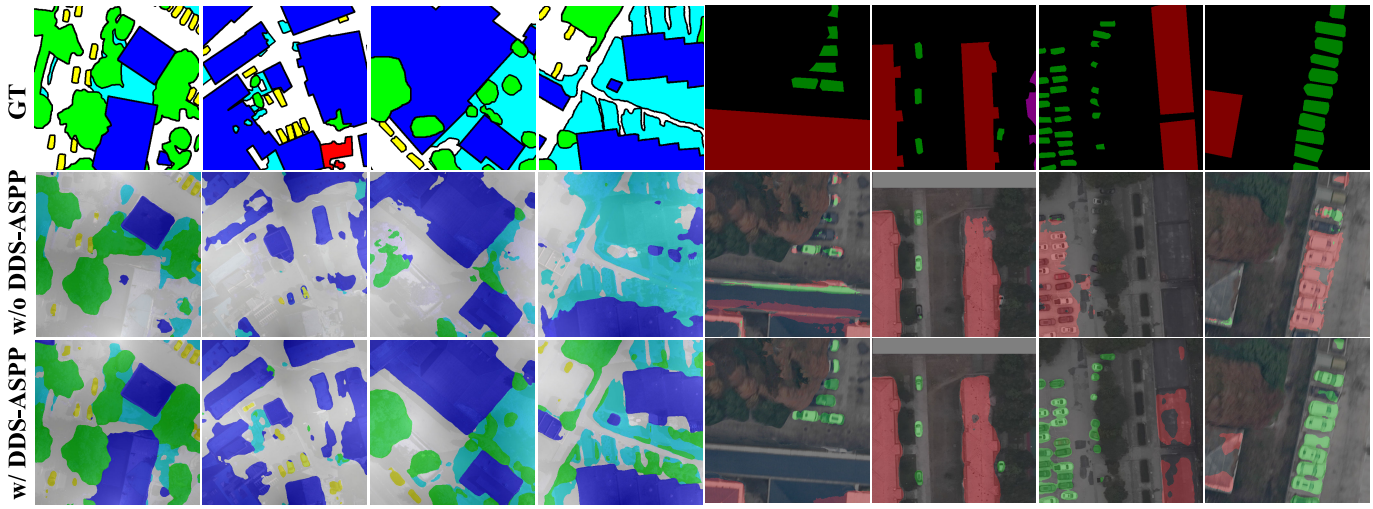


Fig. 19. *w/o* and *w/* means *with* and *without*. In contrast to the model lacking DDS-ASPP, our approach attains comparatively precise car classification. With a large receptive field, DDS-ASPP can grasp content with diverse scales in severe conditions.

To investigate the effect of various dilation rates in DDS-ASPP, we conducted detailed experiments on DDS-ASPP. Table III displays the specific dilation rates within the module, with "()" denoting the values, and max scale referring to the maximum receptive field scale. Two conclusions can be drawn from Table III. First, compared with ASPP, DDS-ASPP improves the performance while reducing the complexity. Second, the performance improves, as the receptive field increases, but reaches a bottleneck when the receptive field size exceeds 127, which is the size of the model output. Fig. 20 [36], [37] depicts the effective receptive fields (ERFs) of different models. According to an aggregated contribution

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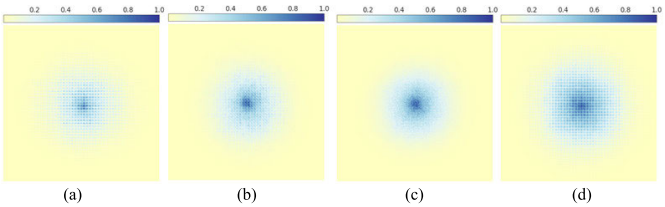


Fig. 20. Larger ERF is represented by a more extensive dark region. ResNet provides limited benefits for enlarging ERF. DDS-ASPP achieves a large ERF, thereby enhancing robustness. (a) ResNet-50. (b) HRNet-W18. (c) HRNet-W48. (d) w/ DDS-ASPP.

TABLE III
PERFORMANCE COMPARISON OF DIFFERENT ASPP STRUCTURES ON FOG CORRUPTED VAIHINGEN DATASET

Structure	GFLOPS	Max Scale	mIoU(%)
ASPP(6,12,18)	11.61	73	55.43
ASPP(6,12,18,24)	15.32	121	57.12
DDS-ASPP(3, 6, 12)	0.975	43	52.66
DDS-ASPP(6, 12, 18, 30)	1.95	133	60.15
DDS-ASPP(3, 6, 12, 18, 24)	4.63	127	62.01
DDS-ASPP(3, 6, 12, 18, 30)	4.51	139	61.96
DDS-ASPP(6, 12, 18, 24, 30)	4.43	181	61.83

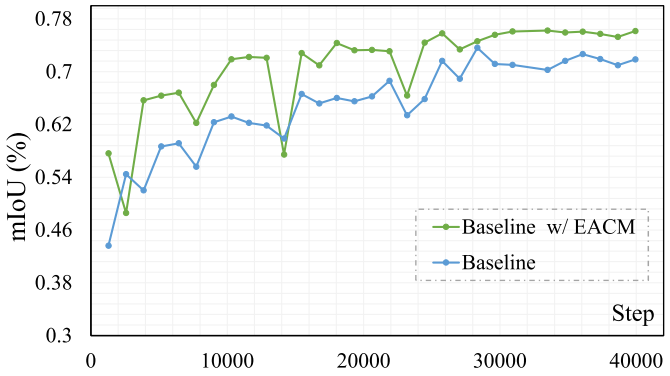


Fig. 21. Comparison of models with and without EACM. The model without EACM (baseline) exhibits a significant decrease in performance.

score matrix, ERF reflects the relative influence of all pixels on the central position of the feature map, which is from the final layer. Compared with the baseline ResNet, HRNet and DDS-ASPP-augmented HRNet possess increasingly larger and more dispersed ERFs, with more pixels impacting the central region. The findings suggest that DDS-ASPP possesses the potential to expand the receptive field for capturing multiscale objects, thereby contributing to the improvement of foggy scene comprehension.

C. Detailed Study on EACM

We conducted a rigorous analysis of EACM. The model training process in Fig. 21 was carried out on the clean Potsdam dataset. We replaced EACM in the original SRE with a standard Conv2D unit as a baseline model for comparison. It is evident that EACM achieved approximately 9% higher than the baseline. Fig. 22 illustrates the performance gains afforded by EACM. Vehicles that are obscured by trees have fewer

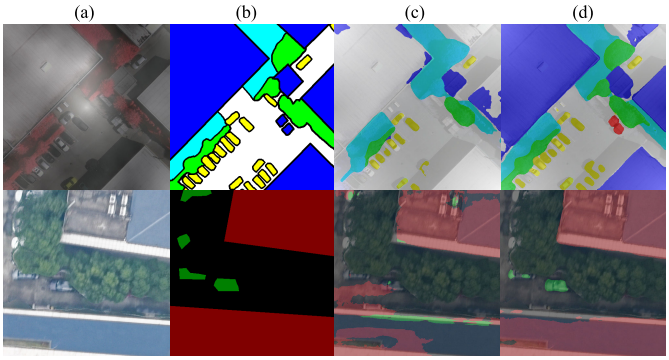


Fig. 22. Comparison of classification performance between models without and with EACM. The addition of EACM significantly improves the extraction of coarse representations. (a) Image. (b) GT. (c) w/o EACM. (d) w/ EACM.

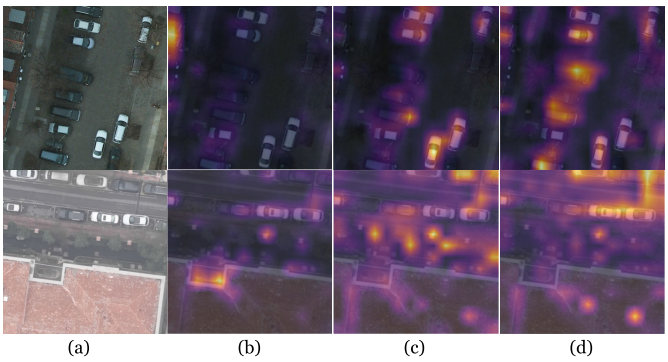


Fig. 23. Attention map is generated for car, where darker colors indicate a higher degree of contribution to the final classification. (a) Image. (b) Baseline. (c) w/ EACM. (d) w/ SRA.

discernible visual features. Model with EACM successfully classified several cars with varying orientations under fog and trees, thus reducing the chance of misclassifying them as buildings.

Fig. 23 shows the heat map generated for car, where darker colors indicate a higher degree of contribution to the final classification. Compared with the baseline, EACM significantly improves the capture of semantic information. EACM extracts a coarse-grained classification, and with the addition of SRA, the model focuses more on the car details while discarding some irrelevant representations.

VI. DISCUSSION

We also conduct a comparison study on Potsdam to evaluate the accuracy and robustness for further research. Methods we select for comparison are mainly SOTA, and they are FCN [6], UNet [12], PSPNet [9], DeepLabV3+ [16], CCNet [20], and OCRNet [21]. Among these networks, FCN, UNet, PSPNet, DeepLabV3+, and CCNet utilize ResNet18 as the backbone architecture, while OCRNet employs HRNetW18 as its backbone.

Fig. 24 reveals that many models struggle to precisely classify cars, and some of them are erroneously identified as road surfaces under fog [such as Fig. 24(d)]. Fig. 24(c) and (f) encompasses significant portions of vegetation that are inaccurately interpreted as background. In contrast to several other models, our approach demonstrates the ability to classify land

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TABLE IV

COMPARISON STUDY RESULT SHOWS DIFFERENT COMPONENTS INTEGRATED FOR THE CLEAN AND FOG CORRUPTED POTSDAM TEST SET. IOU IS AVERAGED ACROSS ALL SEVERITY LEVELS. HIGHEST IOU PER CLASS IS BOLD

Method	GFLOPs	Imp Surf*		Building		Low Veg*		Tree		Car		mIoU(%)		OA(%)	
		Clean	Fog	Clean	Fog	Clean	Fog	Clean	Fog	Clean	Fog	Clean	Fog	Clean	Fog
FCN	50.46	70.59	25.35	71.76	37.99	68.99	6.12	66.09	14.88	41.35	13.51	63.76	19.57	78.05	40.74
UNet	51.95	75.35	36.59	77.14	55.22	70.38	8.91	69.63	36.65	61.64	42.47	70.83	35.97	81.54	46.18
PSPNet	54.26	78.01	35.28	76.98	61.32	73.24	9.61	68.90	38.79	61.24	47.83	71.67	38.57	80.80	47.90
DeepLabV3+	58.71	76.73	37.35	78.85	64.01	72.39	9.72	69.65	40.47	58.47	46.02	71.22	39.51	82.82	49.42
CCNet	49.21	76.76	35.64	75.34	59.58	69.52	8.47	68.85	36.99	61.05	46.63	70.30	37.46	82.67	46.10
OCRNet	53.07	79.12	41.43	85.72	69.26	74.34	11.07	73.69	44.62	76.49	63.19	77.87	45.91	87.37	58.78
Ours	66.64	**83.58**	**45.97**	**88.65**	**73.02**	**72.06**	**11.87**	**77.91**	**48.99**	**80.29**	**66.18**	**80.50**	**49.21**	**89.12**	**61.19**

TABLE V

MCD IN COMPARISON STUDY FOR SEVERAL ARCHITECTURES ON THE POTSDAM DATASET. FCN IS UTILIZED AS THE REFERENCE. IOU IS AVERAGED ACROSS ALL SEVERITY LEVELS. LOWEST IOU PER CLASS IS BOLD

Method	Imp Surf*		Building		Low Veg*		Tree		Car		mIoU(%)		OA(%)	
	Clean	Fog	Clean	Fog	Clean	Fog	Clean	Fog	Clean	Fog	Clean	Fog	Clean	Fog
FCN	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
UNet	83.82	84.94	80.95	72.21	95.52	97.03	89.56	74.42	65.40	66.52	80.49	79.61	84.10	90.82
PSPNet	74.77	86.70	81.52	62.38	86.29	96.28	91.71	71.91	66.09	60.32	78.15	76.38	87.47	87.92
DeepLabV3+	79.12	83.92	74.89	58.04	89.04	96.17	89.50	69.94	70.81	62.41	79.41	75.20	78.27	85.35
CCNet	79.02	86.22	87.32	65.18	98.29	97.50	91.86	74.02	66.41	61.71	81.93	77.75	78.95	90.96
OCRNet	71.00	78.46	50.57	49.57	**82.75**	94.73	77.59	65.06	40.09	42.56	61.05	67.25	57.54	69.56
Ours	**55.83**	**72.38**	**40.19**	**43.51**	90.10	**93.88**	**65.14**	**59.93**	**33.61**	**39.10**	**53.81**	**63.15**	**49.57**	**65.49**

TABLE VI

RCD IN COMPARISON STUDY FOR SEVERAL ARCHITECTURES ON THE POTSDAM DATASET. FCN IS UTILIZED AS THE REFERENCE. IOU IS AVERAGED ACROSS ALL SEVERITY LEVELS. LOWEST IOU PER CLASS IS BOLD

Method	Imp Surf*	Building	Low Veg*	Tree	Car	mIoU(%)	OA(%)
FCN	100.00	100.00	100.00	100.00	100.00	100.00	100.00
UNet	85.68	64.91	97.77	64.40	68.86	78.89	94.77
PSPNet	94.45	46.37	101.21	58.80	**48.17**	74.93	88.18
DeepLabV3+	87.05	**43.94**	99.68	56.98	44.72	71.75	89.52
CCNet	90.89	46.67	97.11	62.21	51.80	74.33	98.02
OCRNet	83.31	48.74	100.64	56.77	47.77	72.33	76.63
Ours	**83.13**	46.28	**95.74**	**56.47**	50.68	**70.82**	**74.86**

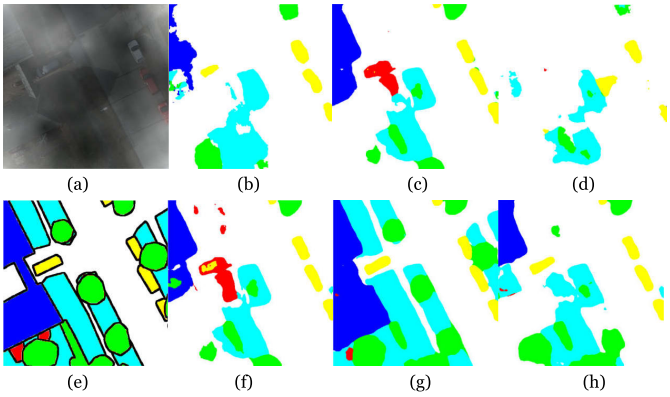


Fig. 24. Qualitative comparisons between robustness performance with respect to fog corruption of level 3. (a) Severity level 3. (b) UNet. (c) PSPNet. (d) CCNet. (e) GT. (f) DeepLabV3+. (g) Ours (clean). (h) Ours (fog).

covers robustly (e.g., severity level 3). The model experiences a relatively minor performance deterioration when compared with normal weather conditions. Nevertheless, there remain certain areas where the model is unable to fully classify the entire rooftop. We also generate the heatmap using SEG-GRAD-CAM [38] for an interpretable visualization (Fig. 25). It enables us to examine which localized regions contribute most significantly to the overall classification. The selected target class is car, and the chosen target layer is the normalized layer in the penultimate block of the network. The deeper the color, the closer the confidence level is to one, suggesting a greater probability that the object will be classified as a car. Occasionally, the surrounding area of a car also exhibits a darker hue, signifying the enhanced ability to assimilate contextual information. The region delineated by dashed lines in Fig. 25(b) proves more challenging to classify accurately due to the presence of dense fog and foliage. As evident in the figure, the model in Fig. 25(d)–(f) misclassifies more extraneous regions as cars, including large rooftops and tree branches. In contrast, the models in Fig. 25(g)–(i) demonstrate greater robustness. It can be observed from Fig. 25 that our

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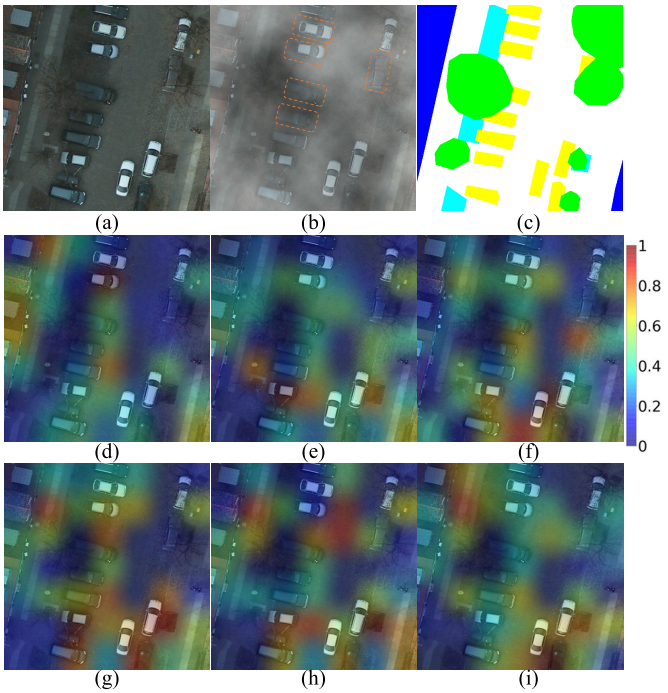


Fig. 25. Heatmap is generated using segmentation gradient-weighted class activation mapping (SEG-GRAD-CAM) for the comparison models with regard to the class car. (a) Clean. (b) Corrupted. (c) GT. (d) FCN. (e) UNet. (f) PSPNet. (g) DeepLabV3. (h) HRNet. (i) HRNet w/ SRE w/ SRA.

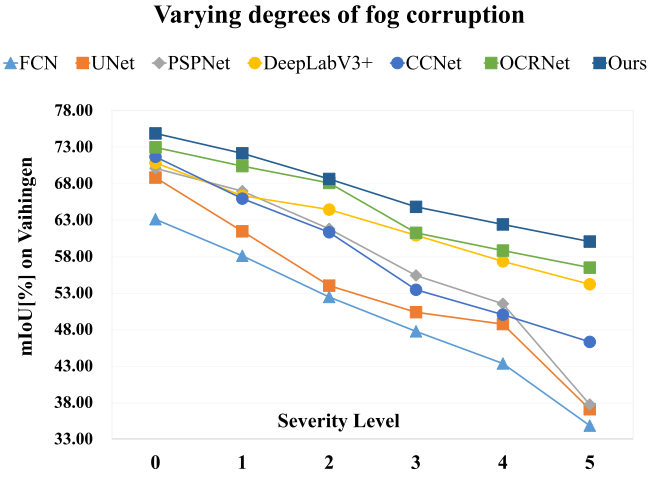


Fig. 26. Line plot is about model performance on Vaihingen with regard to different severity levels of fog. Severity level 0 signifies the clean test set.

TABLE VII
EXPERIMENT ENVIRONMENT

Software	Version	Hardware	Version
CUDA	10.2	CPU	i7-5930K
cuDNN	7.6	GPU	RTX3090ti
Pytorch	1.7	RAM	64GB
Fast.ai	2.2.2	HDD	SSD 2T

model highlights the edge and surrounding pixels of cars. Through acquiring the long-range dependency, our model can capture more informative details of the tiny objects to improve the robustness.

Our model outperforms some SOTA methods on both clean and fog corrupted test set, as seen in the Tables IV–VI. **Imp Surf*** and **Low Veg*** mean impervious surface and low vegetation, respectively. Both mCD and rCD are below 100% compared with FCN, which is selected as the reference. mIoU and OA both show advantage. Compared with the clean dataset, our model has a 31.29% loss in performance on the corrupted test set. There is a gap with regard to the performance on Vaihingen (mIoU loss of 8.34%), probably due to complex scenes and higher degrees of object occlusion under fog. DeepLabV3+ and OCRNet both deliver excellent robustness results. In terms of rCD performance for building, DeepLabV3+ is ahead by 2.34% compared with our model. This indicates a relatively small degradation compared with clean reference. In addition to the primary analysis, we have incorporated a discussion of the computation complexity. As illustrated in Table IV, our proposed model demonstrates an increase in GFLOPs by 32.06% compared with FCN, while OA improved by 50.19%. In contrast, when compared with OCRNet, our model's GFLOPs increased by 25.57%, yet the performance enhancement in OA was a mere 4.1%. Consequently, it can be inferred that there is a strong correlation between robustness and complexity. For future research, it is of great significance to explore strategies that reduce model complexity while simultaneously enhancing its capability to classify land cover in foggy conditions.

As can be seen from quantitative data in Fig. 26, the classification performance of each model falls, as the fog concentration increases. Our model has an improvement in mIoU of 11.76% with regard to the first level corruption and 25.21% at the fifth level compared with FCN. In comparison with OCRNet, our model has a 1.9%–3.57% improvement in mIoU across various fog corruption degrees. Drawing conclusion from Fig. 26, in terms of mCD, our model is robust for objects of diverse sizes.

VII. CONCLUSION

The main work of this article is to improve LCC in foggy conditions. Based on HRNet, we propose SRA and SRE that consists of DDS-ASPP and EACM. Based on a large receptive field, DDS-ASPP can encode multiscale representation, which pertains to various classes efficiently. EACM can generate a feature map associated with pixel representation without adding extra computational load during the inference stage. SRA can capture long-range dependency by modeling the relationship between dual representations. We generate a test set, which is composed of five severities of fog corruption to evaluate the model. Experimental results validate the effectiveness of the proposed method and verify that our model is robust for LCC under fog with less performance degradation. Meanwhile, we also observe that our model is still characterized with insufficient classification for building roofs in dense fog with low efficiency, which will be the focus of our future research.

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APPENDIX

A. Environment

Since the software and hardware version in the experiment will greatly influence the performance about classification accuracy, versions of the major software and hardware used are listed in Table VII.

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